The HAICu Project (WP2): Continual Machine Learning and Humans in the Loop.

Romein, C.A.
info@caromein.nl
UTwente, Netherlands, The; Universität Bern, Switzerland

Wolf, B.J.
ben.wolf@nhlstenden.com
NHL Stenden, the Netherlands

Weggeman, S.J.L.
sjors.weggeman@nhlstenden.com
NHL Stenden, the Netherlands

van Schuijlenburg, K.
k.van.schuijlenburg@rug.nl
University of Groningen, the Netherlands

Dhali, M.A.
m.a.dhali@rug.nl
University of Groningen, the Netherlands

Dijkstra, K.
klaas.dijkstra@nhlstenden.com
NHL Stenden, the Netherlands

Weber, A.
a.weber@utwente.nl
UTwente, Netherlands, The

Schomaker, L.R.B.
l.r.b.schomaker@rug.nl
University of Groningen, the Netherlands

The *digital Humanities Artificial Intelligence and Cultural heritage* (HAICu) project is an emerging interdisciplinary and large-scale research initiative at the intersection of digital humanities, artificial intelligence, and cultural heritage preservation. By applying new computational techniques to analyze handwritten manuscripts with complex layouts, one work package (WP2) of the project aims to transcend traditional methodological boundaries in text language-centered digital humanities research. Supported by the Dutch Research Agenda (NWO-NWA), HAICu exemplifies the synergistic potential of research in the humanities and artificial intelligence. The main aim of the work package is to research and develop a machine-learning architecture for the cultural heritage sector, which promotes continual machine learning through continual, 'life-long' harvesting of labels and annotations to enable multi-modal data mining (Schomaker 2020). The work package's innovations in layout clustering, document structure detection, and contextual text linking represent significant progress toward scalable solutions for studying and interpreting handwritten and other multimodal collections.

Beyond technological advances, HAICu aims to promote digital literacy and enhance historical understanding of multimodal archive collections from three different institutions: the National Archives in The Hague (NA), the Groninger Archives (GA), and the Collection Overijssel (CO). These archives have different levels of existing metadata, creating unique challenges for computational analysis. The project distinguishes between information-rich digitized collections and those with limited digital enrichment (e.g. raw transcriptions, recognized layout elements) and contextual information.

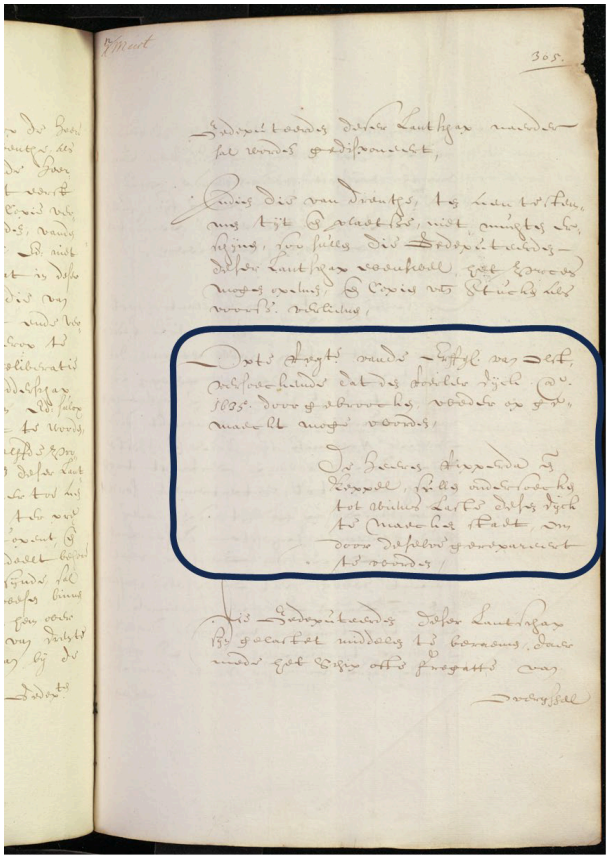

Figure 1 Example of the openTSNE mapping of the Dutch Ministry of Colonial Affairs Archive.

In the context of the National Archives, we work with the archives of the Dutch Ministry of Colonial Affairs (1850-1900), entailing ca. 4 million scans. Researchers from the NHL-Stenden University for Applied Sciences use advanced deep-learning technologies such as Laypa (Klut/ Koert/ Sluijter 2023), DinoV2 vision transformers(Oquab et al 2023), and openTSNE mapping (Poličar et al 2024) to develop sophisticated latent space embeddings. These techniques cluster page layouts based on structural features, going beyond conventional automatic text recognition to capture historical documents' nuanced visual and contextual complexity. Visualizing this latent space exposes clusters of handwriting styles for printed tabular structures, implicit tables on mixed layout pages, and writing density for different column widths.

Researchers from the University of Twente study the Staten van Overijssel collection, comprising 60,000 pages. It documents centuries of political and social interactions in the Netherlands (1578-1795). The project pays particular attention to historical petitions/rekwests, representing the 'voice of the people' and offering insights into societal concerns. One of the core objectives is to generate innovative automatic metadata - likely with ANNIF (Suominen 2019) - that will make these documents more accessible and interpretable for the general public.

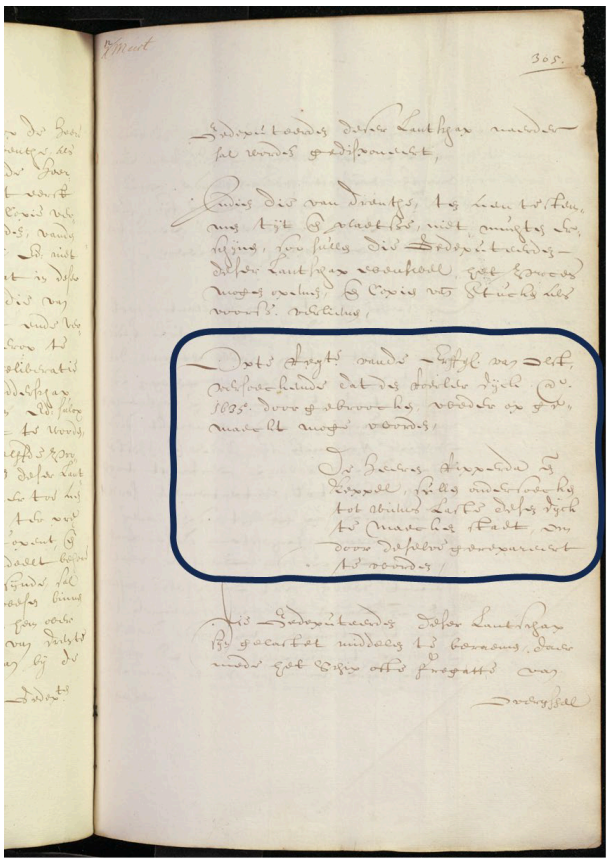

Figure 2 Example from the Overijssel Provincial State archive, indicated is an request and an answer within the minutes.

Central to WP2's methodology is research into a robust human-in-the-loop framework that recognizes the limitations of fully automated systems. Through integrative methods for interpreting multimodal cultural heritage collections, the project bridges technological innovation and historical scholarship. It offers a new approach to understanding our collective past. The project continuously refines machine learning models by integrating academic expertise and volunteer contributions. This collaborative approach democratizes cultural heritage research and actively engages diverse stakeholders in preserving and interpreting historical archives. By combining advanced computational methods with nuanced human interpretation, the project ensures that historical narratives are preserved, dynamically understood, and meaningfully communicated. At its core, HAICu represents a compelling paradigm of interdisciplinary research. It demonstrates how artificial intelligence and digital humanities can work together to transform our engagement with historical archives.

Acknowledgement:

This work is supported by the Dutch Research Agenda (NWA) of the Netherlands Organization for Scientific Research (NWO), grant: NWA.1518.22.105.

Bibliography

Klut, S., van Koert, R. and Sluijter, R., 2023. Laypa: A Novel Framework for Applying Segmentation Networks to Historical Documents. In Proceedings of the 7th International Workshop on Historical Document Imaging and Processing (HIP '23). Association for Computing Machinery, New York, NY, USA, 67–72. https://doi.org/10.1145/3604951.3605520 .

Oquab, M., Darcet, T., Moutakanni, T., Vo, H., Szafraniec, M., Khalidov, V., Fernandez, P., Haziza, D., Massa, F., El-Nouby, A. and Assran, M. , 2023. Dinov2: Learning robust visual features without supervision. arXiv preprint arXiv:2304.07193 . https://doi.org/10.48550/arXiv.2304.07193.

Poličar, P.G., Stražar, M. and Zupan, B. , 2024. openTSNE: A Modular Python Library for t-SNE Dimensionality Reduction and Embedding. Journal of Statistical Software , 109 , pp.1-30. doi: https://doi.org/10.18637/jss.v109.i03 .

Schomaker, L.R.B. (2020). Lifelong Learning for Text Retrieval and Recognition in Historical Handwritten Document Collections, In: Series in Machine Perception and Artificial Intelligence Handwritten Historical Document Analysis, Recognition, and Retrieval — State of the Art and Future Trends, pp. 221-248, Singapore: World Scientific. https://doi.org/10.1142/9789811203244_0012 .

Suominen, O. (2019). Annif: DIY automated subject indexing using multiple algorithms. LIBER Quarterly: The Journal of the Association of European Research Libraries, 29(1), 1-25. https://doi.org/10.18352/lq.10285 .